GENETIC ALGORITHM

GENETIC ALGORITHMS

GENETIC ALGORITHM

Introduction

A biologically inspired model of intelligence

The principles of *biological evolution* are applied to find solutions to difficult problems

The problems are not solved by reasoning logically about them; rather populations of competing candidate solutions are spawned and then evolved to become better solutions through a process patterned after *biological evolution*

GENETIC ALGORITHM

Introduction

Less worthy candidate solutions tend to die out, while those that show promise of solving a problem survive and reproduce by constructing new solutions out of their components

GA begin with a population of candidate problem solutions

Candidate solutions are evaluated according to their ability to solve problem instances: only the fittest survive and combine with each other to produce the next generation of possible solutions

GENETIC ALGORITHM

Introduction

Thus increasingly powerful solutions. Learning is viewed as a competition among a population of evolving candidate problem solutions

This method is heuristic in nature and it was introduced by John Holland in 1975

GENETIC ALGORITHM

Basic Algorithm

```
begin
  set time t = 0;
  initialise population P(t) = {x_1^t, x_2^t, …, x_n^t} of solutions;
  while the termination condition is not met do
    begin
      evaluate fitness of each member of P(t);
      select some members of P(t) for creating offspring;
      produce offspring by genetic operators;
      replace some members with the new offspring;
      set time t = t + 1;
    end
end
```

GENETIC ALGORITHM

Chromosome

GENETIC ALGORITHM

Representation of Solutions: The Chromosome

Gene*: A basic unit, which represents one characteristic of the individual. The value of each gene is called an *allele

***Chromosome*: A string of genes; it represents an individual i.e. a possible solution of a problem. Each chromosome represents a point in the search space**

***Population*: A collection of chromosomes**

An appropriate chromosome representation is important for the efficiency and complexity of the GA

GENETIC ALGORITHM

Representation of Solutions: The Chromosome

The classical representation scheme for chromosomes is binary value of fixed length. However, real valued chromosomes are also possible

In binary coded chromosomes, every gene has two alleles
In real coded chromosomes, a gene can be assigned any value from a domain of values

In the case of an *I*-dimensional search space, each chromosome consists of *I* variables with each variable encoded as a bit string

GENETIC ALGORITHM

Representation of Solutions: The Chromosome

Example: Cookies Problem

Two parameters sugar and flour (in kgs). The range for both is 0 to 9 kgs. Therefore a chromosome will comprise of two genes called sugar and flour

5	**1**

Chromosome # 01

2	**4**

Chromosome # 02

GENETIC ALGORITHM

Representation of Solutions: The Chromosome

Example: Model Learning

Use GA to learn the concept *Yes Reaction* from the Food Allergy problem's data

Number	Restaurant	Meal	Day	Cost	Reaction
1	Sam's	breakfast	Friday	cheap	yes
2	Lobdell	lunch	Friday	expensive	no
3	Sam's	lunch	Saturday	cheap	yes
4	Sarah's	breakfast	Sunday	cheap	no
5	Sam's	breakfast	Sunday	expensive	no

GENETIC ALGORITHM

Example: Model Learning

A potential model of the data can be represented as a chromosome with the genetic representation:

Gene # 1 **Restaurant**	**Gene # 2** **Meal**	**Gene # 3** **Day**	**Gene # 4** **Cost**

The alleles of genes are:

Restaurant gene: Sam, Lobdell, Sarah
Meal gene: breakfast, lunch
Day gene: Friday, Saturday, Sunday
Cost gene: cheap, expensive

GENETIC ALGORITHM

Representation of Solutions: The Chromosome

Rules Representation

Rules are often represented by bit strings (because they can be easily manipulated by genetic operators), but other numerical and symbolic representations are also possible

Set of if-then rules:

Specific sub-strings are allocated for encoding each rule pre-condition and post-condition

Example: Suppose we have an attribute "Outlook" which can take on values: *Sunny*, *Overcast* or *Rain*

GENETIC ALGORITHM

Representation of Solutions: The Chromosome

Rules Representation

We can represent it with 3 bits:

100 would mean the value *Sunny*,
010 would mean *Overcast* &
001 would mean *Rain*

110 would mean *Sunny* or *Overcast*
111 would mean that we don't care about its value

The pre-conditions & post-conditions of a rule are encoding by concatenating the individual representation of attributes

GENETIC ALGORITHM

Representation of Solutions: The Chromosome

Rules Representation

Example:

If (Outlook = Overcast or Rain) and Wind = strong
then Play Tennis = No

can be encoded as 010001

GENETIC ALGORITHM

Reproduction Operators

GENETIC ALGORITHM

Reproduction Operators

Genetic operators are applied to chromosomes that are selected to be parents, to create offspring

Basically of two types: Crossover and Mutation

Crossover operator create offspring by recombining the chromosomes of selected parents

Mutation is used to make small random changes to a chromosome in an effort to add diversity to the population

GENETIC ALGORITHM

Reproduction Operators: Crossover

Crossover operation takes two candidate solutions and divides them, swapping components to produce two new candidates

GENETIC ALGORITHM

Reproduction Operators: Crossover

Figure illustrates crossover on bit string patterns of length 8

The operator splits them and forms two children whose initial segment comes from one parent and whose tail comes from the other

Input Bit Strings

1 1 # 0 1 0 1 # **# 1 1 0 # 0 # 1**

Resulting Strings

1 1 # 0 # 0 # 1 **# 1 1 0 1 0 1 #**

GENETIC ALGORITHM

Reproduction Operators: Crossover

Two genes sugar and flour (in kgs)
Crossover operation on chromosomes

5	**1**
2	**4**

→

5	**4**
2	**1**

GENETIC ALGORITHM

Reproduction Operators: Crossover

The place of split in the candidate solution is an arbitrary choice. This split may be at any point in the solution

This splitting point may be randomly chosen or changed systematically during the solution process

Crossover can unite an individual that is doing well in one dimension with another individual that is doing well in the other dimension

GENETIC ALGORITHM

Reproduction Operators: Crossover

Two types: Single point crossover & Uniform crossover

Single type crossover

This operator takes two parents and randomly selects a single point between two genes to cut both chromosomes into two parts (this point is called cut point)

The first part of the first parent is combined with the second part of the second parent to create the first child

The first part of the second parent is combined with the second part of first parent to create the second child

1000010 1000001

1110001 1110010

GENETIC ALGORITHM

Reproduction Operators: Crossover

Uniform crossover

The value of each gene of an offspring's chromosome is randomly taken from either parent

This is equivalent to multiple point crossover

1000010

1110001 1010010

GENETIC ALGORITHM

Reproduction Operators: Mutation

Mutation is another important genetic operator

Mutation takes a single candidate and randomly changes some aspect (gene) of it

For example, mutation may randomly select a bit in the pattern and change it, switching a 1 to a 0 or to # (don't care)

GENETIC ALGORITHM

Reproduction Operators: Mutation

Mutation is important in that the initial population may exclude an essential component of a solution

For example, if no member of the initial population has a 1 in the first position, then crossover in the middle, cannot produce a child that could become a solution

Mutation Rate = 0.01

GENETIC ALGORITHM

Selection Process

GENETIC ALGORITHM

Selection Process

It is used to select parents from the current population. The selection is primarily based on the fitness. The better the fitness of a chromosome, the greater its chance of being selected to be a parent

The rate at which a selection algorithm selects individuals with higher fitness is *selective pressure*

If there is not enough *selective pressure*, the population will fail to converge upon a solution. If there is too much, the population may not have enough diversity & converge prematurely

GENETIC ALGORITHM

Selection Process: Random Selection

Random Selection:

Individuals are selected randomly with no reference to fitness at all

All the individuals, good or bad, have an equal chance of being selected

GENETIC ALGORITHM

Selection Process: Proportional Selection

Proportional Selection:

We can select the fittest chromosomes
However, the selection of only the fitter chromosomes may result in the loss of a correct gene value which may be present in a less fit member

One way to overcome this risk is to assign probability of selection to each chromosome based on its fitness

In this way even the less fit members have some chance of surviving into the next generation

GENETIC ALGORITHM

Selection Process: Proportional Selection

The probability of selection of a chromosome "i" may be calculated as

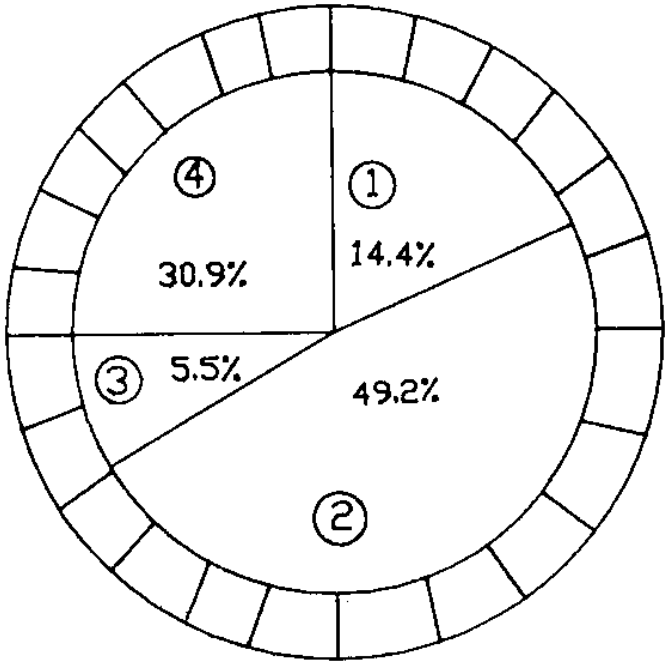
$$p_i = \text{fitness}_i / \sum_j \text{fitness}_j$$

Example

Chromosome	Fitness	Selection Probability
1	7	7/14
2	4	4/14
3	2	2/14
4	1	1/14

GENETIC ALGORITHM

Selection Process: Proportional Selection (Roulette Wheel)



GENETIC ALGORITHM

Selection Process: Proportional Selection

Chromosomes are selected based on their fitness relative to the fitness of all other chromosomes

For this all the fitness are added to form a sum S and each chromosome is assigned a relative fitness (which is its fitness divided by the total fitness S)

A process similar to spinning a roulette wheel is adopted to choose a parent; the better a chromosome's relative fitness, the higher its chances of selection

GENETIC ALGORITHM

Selection Process: Proportional Selection

Once a parent is selected, the wheel is given a spin for finding the second parent. If the same chromosome is selected as the second parent, it is rejected and the wheel is spun again

After finding a pair, a second pair is selected, and so on

A chromosome may get selected several times and appear as a parent several times

GENETIC ALGORITHM

Selection Process: Proportional Selection

Advantage

Selective pressure varies with the distribution of fitness within a population. If there is a lot of fitness difference between the more fit and less fit chromosomes, then the selective pressure will be higher

Disadvantage

As the population converges upon a solution, the selective pressure decreases, which may hinder the GA to find better solutions

GENETIC ALGORITHM

Selection Process: Tournament Selection

Tournament Selection:

One parent is selected by comparing a subset *b* of the available chromosomes, and selecting the fittest; a second parent may be selected by repeating the process

The selection pressure increases as *b* increases.

Value of *b* = 2 is most commonly used

GENETIC ALGORITHM

Selection Process: Tournament Selection

Its advantage is that the worse individuals of the population will have very little probability of selection, whereas the best individuals will not dominate the selection process, thus ensuring diversity

GENETIC ALGORITHM

Selection Process: Rank based selection

Rank Based Selection:

Rank based selection uses the rank ordering of the fitness values to determine the probability of selection and not the fitness values themselves

This means that the selection probability is independent of the actual fitness value

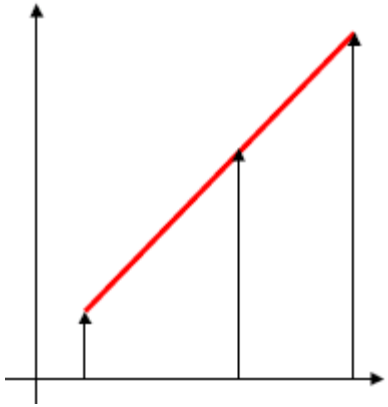
Ranking therefore has the advantage that a highly fit individual will not dominate in the selection process as a function of the magnitude of its fitness

GENETIC ALGORITHM

Selection Process: Rank based selection

The population is sorted from best to worst according to the fitness

Each chromosome is then assigned a new fitness based on a *linear* ranking function



New Fitness = (P – r) + 1

where P = population size, r = fitness rank of the chromosome

If P = 11, then a chromosome of rank 1 will have a New Fitness of 10 + 1 = 11 & a chromosome of rank 6 will have 6

GENETIC ALGORITHM

Selection Process: Rank based selection

A user adjusted slope can also be incorporated

$$\text{New Fitness} = \{(P - r)\ (\max - \min)/(P - 1)\} + \min$$

where max and min are set by the user to determine the slope (max - min)/(P – 1) of the function

Let P = 11, max = 8, min = 3,
then a chromosome of rank 1 will have a New fitness of 10*5/10 + 3 = 8
& a chromosome of rank 6 will have 5*5/10 + 3 = 5.5

GENETIC ALGORITHM

Selection Process: Rank based selection

Once the new fitness is assigned, parents are selected by the same roulette wheel procedure used in proportionate selection

Advantage: selective pressure, once determined by the user, remains constant

Disadvantage:

- **Population must be sorted**
- **Two Chromosomes with the same fitness will not have the same probability of being selected**

GENETIC ALGORITHM

Selection Process: Survival of the most diverse

Diversity

If a chromosome is close to the best chromosome and has a higher score than a chromosome that is far away (diverse) but has a slightly lower score, then the far away chromosome will be wiped out

This results in uniformity, even in large populations

GENETIC ALGORITHM

Selection Process: Survival of the most diverse

Diversity of a chromosome from the already selected chromosomes can be calculated as

- **Take the distance of the chromosome from a selected chromosome**
- **Square the distance (to remove negative distance)**
- **Sum all squared distances (from selected chromosomes)**
- **The larger this sum the more diverse is the chromosome**

$$\text{Diversity} = \sum_i \text{distances}_j^2$$

One way to incorporate diversity is to rank the chromosomes by the sum of fitness rank and diversity rank

GENETIC ALGORITHM

Selection Process: Survival of the most diverse

Hence diversity (or being different) can be incorporated into the selection procedure and a chromosome which is almost as good as another chromosome but is far away from the already selected chromosomes has a better chance of survival

GENETIC ALGORITHM

Next Generation

GENETIC ALGORITHM

The New Generation

The new offspring can replace the old population without any fitness comparison

or

only the better ones from the new and old make it to the new generation (more processing needed)

GENETIC ALGORITHM

New Generation: Elitism

Elitism is a value between 0 and 1, which represents the fraction of the individuals of a population that will be duplicated to the next generation

Example P = 20 and elitism = 0.1, then 2 individuals of current population do not get replaced

The elite chromosome selection may be based on highest fitness values. If highest fitness valued chromosomes are carried over to the next generation, we ensure that the maximum fitness does not decrease from one generation to next

GENETIC ALGORITHM

New Generation: Number of Duplicates Allowed

Duplicates are Chromosomes that are same

If they are allowed then that chromosome has higher probability of producing an offspring, and may probably create many offspring

GENETIC ALGORITHM

New Generation: Number of Duplicates Allowed

Eliminating duplicates increases the efficiency of the genetic search and reduces the danger of premature convergence

Eliminating duplicates means that if an offspring is created which is a duplicate of a chromosome of the current population, we terminate it immediately and create a new one. It increases the processing time in large populations

GENETIC ALGORITHM

Termination Requirement

GENETIC ALGORITHM

Termination Requirement

The GA continues until some termination requirement is met, such as

- **having a solution whose fitness exceeds some threshold**
- **pre-specified number of generations have evolved**
- **the fitness of solutions becomes stable & stops improving**

GENETIC ALGORITHM

Population Size

GENETIC ALGORITHM

Population Size

Number of individuals present in an iteration (generation)

If the population size is too large, the processing time is high and the GA tends to take longer to converge upon a solution (because less fit members have to be selected to make up the required population)

If the population size is too small, the GA is in danger of premature convergence upon a sub-optimal solution (all chromosomes will soon have identical traits). This is primarily because there may not be enough diversity in the population to allow the GA to escape local optima